

PATENT RESEARCH REPORT

1. ABSTRACT

This technical report synthesizes patent evidence for the preparation, polymerization, and copolymerization of Polycarbonate. Analyzing 15 primary patents, the report identifies key technical approaches including continuous melt polymerization, transition-metal-catalyzed copolymerization of carbon dioxide with epoxides (using cobalt-salen and bimetallic zinc catalysts), metal-free anionic copolymerization using phosphazene bases, and the synthesis of functional polycarbonates containing azido groups for biomedical hydrogels. Key process trends highlight a shift toward highly regioselective catalysts, metal-free systems for color-free solids, and precise control of end-group functionality (achieving up to 99% OH end groups) for polyol applications.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

Patent Details - Patent Number: EP2052009 - Patent Title: Method of preparing polycarbonate - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2012 - Material Type: Polycarbonate Terpolymer - Relevance to target: Discloses a method of preparing polycarbonate terpolymers using monomers like BPA, hydroquinone, and methyl hydroquinone with PCP chainstoppers.

Preparation Method - BPA: 33 % (Monomer ratio in terpolymer) - Hydroquinone: 34 % (Monomer ratio in terpolymer) - Methyl hydroquinone: 33 % (Monomer ratio in terpolymer) - PCP: 70 mol% (Chainstopper end groups) - Temperature Zone 1: 280 °C (Zone 1 Temperature) - Temperature Zone 2: 290 °C (Zone 2 Temperature) - Temperature Zone 3: 300 °C (Zone 3 Temperature) - Ultrasonic Treatment: 15 min (Ultrasonic bath treatment time)

Relevant Experimental Evidence - Terpolymer synthesis with 33% BPA, 34% hydroquinone, and 33% methyl hydroquinone. - Addition of para-cumyl phenol (PCP) chainstopper to yield 30, 50, or 70 mol% end groups at a target molecular weight of 30,000 g/mol. - Thermal processing zones set at 280°C, 290°C, and 300°C. - Ultrasonic treatment of the terpolymer for 15 minutes followed by shaking for 2 hours.

Technical Relevance Discloses a method of preparing polycarbonate terpolymers using monomers like BPA, hydroquinone, and methyl hydroquinone with PCP chainstoppers.

Patent 2

Patent Details - Patent Number: US8445624 - Patent Title: Processes for producing polycarbonate - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2013 - Material Type: Polycarbonate Copolymer - Relevance to target: Discloses continuous polymerization processes using vertical and horizontal stirring reactors to produce polycarbonates from isosorbide, tricyclodecanedimethanol, and diphenyl carbonate.

Polymerization Method - Residence Time: 60 minutes (Average residence time in reactor) - Isosorbide: 26.6 parts by weight (Isosorbide monomer loading) - Tricyclodecanedimethanol: 15.3 parts by weight (Tricyclodecanedimethanol monomer loading) - Diphenyl Carbonate: 58.0 parts by weight (Diphenyl carbonate monomer loading) - Cesium Carbonate Catalyst: 0.2 % (Cesium carbonate catalyst concentration) - Pressure: 13.3 kPa (Pressure reduction target) - Reaction Time: 4 hours (Total reaction time)

Relevant Experimental Evidence - Continuous feed of molten starting materials through a pipe heated to 140°C into a vertical stirring reactor. - Polymerization initiated using cesium carbonate catalyst (0.2% aqueous solution). - Average residence time maintained at 60 minutes by regulating the polymer discharge line valve. - Resulting polycarbonates exhibit reduced viscosities of 0.55-0.60, T_g of 129°C, and 5% weight loss temperatures of 337-350°C.

Technical Relevance Discloses continuous polymerization processes using vertical and horizontal stirring reactors to produce polycarbonates from isosorbide, tricyclodecanedimethanol, and diphenyl carbonate.

Patent 3

Patent Details - Patent Number: US9771453 - Patent Title: Process of producing polycarbonate by copolymerization of carbon dioxide and ... - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2017 - Material Type: Aliphatic Polycarbonate - Relevance to target: Discloses copolymerization of carbon dioxide and epoxides (like propylene oxide) using cobalt-salen complexes to produce polycarbonates.

Copolymerization Method - CO₂ Pressure: 18 bar (Initial carbon dioxide pressure) - Propylene Oxide: 10.0 g (Propylene oxide monomer loading) - Temperature: 70-75 °C (Polymerization temperature) - Reaction Time: 60 minutes (Polymerization reaction time) - Monomer/Catalyst Ratio: 100,000 ratio (Monomer to catalyst ratio)

Relevant Experimental Evidence - Copolymerization of carbon dioxide and propylene oxide in a 50 mL bomb reactor. - Catalyst complexes prepared using cobalt acetate and various salen ligands with high yields (up to 100%). - Continuous injection of carbon dioxide at 20 bar for 1 hour at 70-75°C. - Induction times vary significantly from 1 hour (dry winter conditions) to 12 hours (high-temperature, high-humidity summer conditions).

Technical Relevance Discloses copolymerization of carbon dioxide and epoxides (like propylene oxide) using cobalt-salen complexes to produce polycarbonates.

Patent 4

Patent Details - Patent Number: US9006347 - Patent Title: Method of synthesising polycarbonates in the presence of a bimetallic catalyst ... - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2015 - Material Type: Polycarbonate Polyol - Relevance to target: Discloses synthesis of polycarbonates by copolymerizing carbon dioxide and cyclohexene oxide (CHO) using bimetallic zinc catalysts and chain transfer agents.

Synthesis Method - Catalyst Loading: 0.1 mol % (Zinc catalyst loading) - Temperature: 80 °C (Reaction temperature) - Stirring Speed: 700 rpm (Stirring speed) - Chain Transfer Agent: 0.4 mol % (Chain transfer agent loading) - Cyclohexene Oxide: 5 ml (Cyclohexene oxide volume) - CO₂ Pressure: 1 atm (Carbon dioxide pressure) - Reaction Time: 1-24 hours (Process duration)

Relevant Experimental Evidence - Copolymerization of CO₂ and cyclohexene oxide (CHO) using bimetallic zinc catalysts ([L1Zn₂(OAc)₂] or [L1Zn₂(CF₃COO)₂]). - Addition of chain transfer agents (A through Q) or water to produce polycyclohexene oxide carbonate polyols and diols. - High selectivity for copolymer formation (>99% carbonate linkages) maintained even in the presence of excess water. - Reaction mixtures quenched using a 5% solution of HCl/MeOH in dichloromethane.

Technical Relevance Discloses synthesis of polycarbonates by copolymerizing carbon dioxide and cyclohexene oxide (CHO) using bimetallic zinc catalysts and chain transfer agents.

Patent 5

Patent Details - Patent Number: EP2305734 - Patent Title: Polycarbonate copolymer and method of producing the same - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2013 - Material Type: Polycarbonate Copolymer - Relevance to target: Discloses polycarbonate copolymers containing isosorbide, 1,4-CHDM, TCDDM, and other dihydroxy compounds with high thermal stability.

Preparation Method - 5% Weight Reduction Temp: 30-56 °C (Temperature at 5% weight loss)

Relevant Experimental Evidence - Synthesis of polycarbonate copolymers using isosorbide, tricyclodecanedimethanol (TCDDM), and other dihydroxy compounds. - Evaluation of thermal properties, including 5% weight reduction temperatures. - Comparison with commercial homopolymers (e.g., lupilon H4000) and specific ternary copolymer compositions.

Technical Relevance Discloses polycarbonate copolymers containing isosorbide, 1,4-CHDM, TCDDM, and other dihydroxy compounds with high thermal stability.

Patent 6

Patent Details - Patent Number: US10865275 - Patent Title: Copolymerization of carbon dioxide and cyclic monomers to form polycarbonates - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2020 - Material Type: Aliphatic Polycarbonate - Relevance to target: Discloses

metal-free anionic copolymerization of carbon dioxide and cyclic monomers (epoxides) using phosphazene bases (t-BuP4) and diols.

Copolymerization Method - CO₂ Pressure: 10 bar (Carbon dioxide pressure) - Reaction Time: 10 hr (Polymerization reaction time) - Reactor Volume: 50 mL (Parr reactor volume) - THF Volume: 1.0 mL (Tetrahydrofuran solvent volume)

Relevant Experimental Evidence - Metal-free anionic copolymerization of epoxides (propylene oxide, cyclohexene oxide, etc.) with CO₂. - Use of 1,4-benzylidimethanol (BDM) and t-BuP4 base to initiate perfectly alternating copolymerization. - Achieved molar masses up to 250,000 g/mol with narrow molecular weight distributions ($M_w/M_n < 1.3$). - Reactions quenched using HCl in methanol (1 mol/L) or 10% HCl.

Technical Relevance Discloses metal-free anionic copolymerization of carbon dioxide and cyclic monomers (epoxides) using phosphazene bases (t-BuP4) and diols.

Patent 7

Patent Details - Patent Number: US12240978 - Patent Title: Methods for producing polycarbonate copolymer and polysiloxane compound, ... - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2025 - Material Type: Polycarbonate-Polysiloxane Copolymer - Relevance to target: Discloses preparation of polycarbonate-polysiloxane copolymers and polyarylene siloxanes.

Preparation Method - Heptane: 20 g (Heptane added to reaction mixture) - Temperature: 90 °C (Hot filtration temperature) - Pressure Reduction Time: 45 hours (Pressure reduction duration) - Molar Yield: 81.1 % (Molar yield of white powders)

Relevant Experimental Evidence - Solidified reaction mixture treated with 20 g heptane, heated to 90°C, and subjected to hot filtration. - Crystals washed with cold heptane (5°C) and dried under reduced pressure (1 hPa) for 45 hours to yield white powders. - Colorless and transparent polyarylene siloxane obtained by heating under normal pressure over 1.5 hours.

Technical Relevance Discloses preparation of polycarbonate-polysiloxane copolymers and polyarylene siloxanes.

Patent 8

Patent Details - Patent Number: EP4011934 - Patent Title: Polyether polycarbonate diol and method for producing same - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2025 - Material Type: Polyether Polycarbonate Diol - Relevance to target: Discloses polyether polycarbonate diols and methods for their production.

Technical Method

Relevant Experimental Evidence - Discloses synthesis and characterization of polyether polycarbonate diols. - Focuses on achieving specific diol structures for polyurethane applications.

Technical Relevance Discloses polyether polycarbonate diols and methods for their production.

Patent 9

Patent Details - Patent Number: US10597498 - Patent Title: Method of producing crystalline polycarbonate powders - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2020 - Material Type: Crystalline Polycarbonate Powder - Relevance to target: Discloses a method for preparing partially crystalline polycarbonate powders with controlled particle size via precipitation.

Preparation Method - Polycarbonate Concentration: 20 wt % (Polycarbonate concentration in DCM) - Acetone Volume: 400 ml (Acetone non-solvent volume) - Stirring Speed: 6000 rpm (High shear mixing speed) - Crystallinity: 20-35 % (Resulting powder crystallinity) - GPC Flow Rate: 1.5 ml/min (GPC flow rate)

Relevant Experimental Evidence - Amorphous polycarbonate dissolved in dichloromethane (20 wt%) and gradually added to acetone (non-solvent) under high shear mixing (6000 rpm). - Resulting powder consists of ~93% particles with an average size of less than 150 μm . - Precipitated polycarbonate exhibits crystallinity between 20% and 35%.

Technical Relevance Discloses a method for preparing partially crystalline polycarbonate powders with controlled particle size via precipitation.

Patent 10

Patent Details - Patent Number: US7723256 - Patent Title: Polycarbonates made using highly selective catalysts - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2010 - Material Type: Aliphatic Polycarbonate - Relevance to target: Discloses highly regioselective copolymerization of carbon dioxide and propylene oxide using cobalt-salen catalysts.

Copolymerization Method - CO₂ Pressure: 10-850 psi (Carbon dioxide pressure range) - Reaction Temperature: 20-25 °C (Reaction temperature range) - Reaction Time: 0.5-4 hours (Reaction time range) - Catalyst Ratio: 400:1 to 600:1 ratio (Alkylene oxide to catalyst ratio) - Carbonate Linkages: 98 % (Carbonate linkages in product)

Relevant Experimental Evidence - Copolymerization of neat propylene oxide and CO₂ in a Parr autoclave using [(R,R)-(salen)CoOAc] or [(R,R)-(salen)CoBr] catalysts. - Achieved high carbonate linkages ($\geq 98\%$) and high regioselectivity (80-85% head-to-tail). - Precipitation of yellow crude complex by adding water (150 mL) to the reaction mixture.

Technical Relevance Discloses highly regioselective copolymerization of carbon dioxide and propylene oxide using cobalt-salen catalysts.

Patent 11

Patent Details - Patent Number: US9868816 - Patent Title: Aliphatic polycarbonate quench method - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2018 - Material Type: Aliphatic Polycarbonate - Relevance to target: Discloses a method for quenching polymerization reactions of aliphatic polycarbonates using specific solvent mixtures.

Technical Method - Propyl Acetate: 50 wt % (Propyl acetate in dope) - Propylene Oxide: 30 wt % (Propylene oxide in dope) - PPC Polymer: 20 wt % (PPC polymer in dope)

Relevant Experimental Evidence - Preparation of poly(propylene carbonate) (PPC) polymer in a pressurized reactor. - Dilution of the reaction mixture (dope) with propyl acetate to yield a composition of 50 wt% propyl acetate, 30 wt% propylene oxide, and 20 wt% PPC polymer. - Quenching process designed to stabilize the polymer and prevent degradation.

Technical Relevance Discloses a method for quenching polymerization reactions of aliphatic polycarbonates using specific solvent mixtures.

Patent 12

Patent Details - Patent Number: US10752614 - Patent Title: Monomers and polymers for functional polycarbonates and poly(ester-carbonates) ... - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2020 - Material Type: Functional Polycarbonate - Relevance to target: Discloses functional cyclic carbonate monomers (like AzDXO) and their ring-opening polymerization to form functional polycarbonates.

Polymerization Method - Monomer Conversion: 99 % (Monomer conversion rate) - Overall Yield: 45.6 % (Overall yield of AzDXO) - Gellation Time: 1 min (Gellation time of hydrogel) - FBS Concentration: 20 % (FBS in expansion media)

Relevant Experimental Evidence - Synthesis of azido-functionalized cyclic carbonate monomer (AzDXO) in two steps with 45.6% overall yield. - Ring-opening polymerization of AzDXO and copolymerization with L-lactide using DBU catalyst. - Rapid gellation (<1 min) of cell-hydrogel constructs via strain-promoted azide-alkyne cycloaddition (SPAAC) click reaction.

Technical Relevance Discloses functional cyclic carbonate monomers (like AzDXO) and their ring-opening polymerization to form functional polycarbonates.

Patent 13

Patent Details - Patent Number: EP2678326 - Patent Title: Monomers and polymers for functional polycarbonates and poly (ester-carbonates) ... - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2017 - Material Type: Functional Polycarbonate - Relevance to target: Discloses functional cyclic carbonate monomers (AzDXO) and polycarbonates for biomedical applications, detailing ROP kinetics.

Polymerization Method - ROP Rate Constant (AzDXO): 0.084 min⁻¹ (ROP rate constant of AzDXO) - ROP Rate Constant (L-LA): 0.116 min⁻¹ (ROP rate constant of L-LA) - Melting Point:

86.31 to 98.43 °C (Melting point range of polymer) - Thermal Stability: 130 °C (Thermal stability limit)

Relevant Experimental Evidence - Determination of ring-opening polymerization (ROP) rate constants for AzDXO (0.084 min⁻¹) and L-lactide (0.116 min⁻¹). - Synthesis of PEG-P(AzDXO) macromers with >95% overall yield. - Characterization of polymer crystallinity and glass transitions using differential scanning calorimetry (DSC).

Technical Relevance Discloses functional cyclic carbonate monomers (AzDXO) and polycarbonates for biomedical applications, detailing ROP kinetics.

Patent 14

Patent Details - Patent Number: US11535706 - Patent Title: Polycarbonate polyol compositions and methods - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2022 - Material Type: Polycarbonate Polyol - Relevance to target: Discloses polycarbonate polyols with high OH end-group content prepared using cobalt catalysts and chain transfer agents.

Synthesis Method - Catalyst Loading: 11.5 mg (Cobalt catalyst loading) - PPN+Cl⁻ Loading: 9.2 mg (PPN+Cl⁻ co-catalyst loading) - Propylene Oxide: 20 ml (Propylene oxide volume) - OH End Groups: 99 % (OH end groups in polyol)

Relevant Experimental Evidence - Synthesis of polycarbonate polyols in a Fisher-Potter bottle or glass vessel under vacuum. - Copolymerization of propylene oxide and CO₂ using cobalt catalysts (E1 or E2) and PPN+Cl⁻ co-catalyst. - Use of chain transfer agents like 1,4-cyclohexanedimethanol, propoxylated pentaerythritol, succinic acid, or adipic acid to control molecular weight. - Resulting polyols contain 97% to 99% OH end groups.

Technical Relevance Discloses polycarbonate polyols with high OH end-group content prepared using cobalt catalysts and chain transfer agents.

Patent 15

Patent Details - Patent Number: US9809678 - Patent Title: Polycarbonate polyol compositions and methods - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2017 - Material Type: Polycarbonate Polyol - Relevance to target: Discloses polycarbonate polyols with high OH end-group content prepared using cobalt catalysts and chain transfer agents (similar to US11535706).

Synthesis Method - Catalyst Loading: 11.5 mg (Cobalt catalyst loading) - PPN+Cl⁻ Loading: 9.2 mg (PPN+Cl⁻ co-catalyst loading) - Propylene Oxide: 20 ml (Propylene oxide volume) - OH End Groups: 99 % (OH end groups in polyol)

Relevant Experimental Evidence - Copolymerization of propylene oxide and CO₂ using cobalt catalysts (E1 or E2) and PPN+Cl⁻ co-catalyst. - Incorporation of starter diols/triols (e.g., 1,4-cyclohexanedimethanol, propoxylated pentaerythritol) to yield polyols with 97-99% OH end

groups. - Reactions quenched with acid, diluted with acetone, and concentrated to yield crude polymer.

Technical Relevance Discloses polycarbonate polyols with high OH end-group content prepared using cobalt catalysts and chain transfer agents (similar to US11535706).

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

Comparison Table: | Patent | CO2 Pressure | Key Finding | |---|---|---| | EP2052009 | Not disclosed | | | US8445624 | Not disclosed | | | US9771453 | Not disclosed | | | US9006347 | Not disclosed | | | EP2305734 | Not disclosed | | | US10865275 | Not disclosed | | | US12240978 | Not disclosed | | | EP4011934 | Not disclosed | | | US10597498 | Not disclosed | | | US7723256 | Not disclosed | | | US9868816 | Not disclosed | | | US10752614 | Not disclosed | | | EP2678326 | Not disclosed | | | US11535706 | Not disclosed | | | US9809678 | Not disclosed | |

Trend Analysis: - Transition-metal catalysts (cobalt-salen and bimetallic zinc) enable highly selective CO2/epoxide copolymerization at moderate temperatures (20-80 °C) but leave metal residues. - Metal-free anionic systems utilizing phosphazene bases (t-BuP4) or PPNCl/triethylborane complexes offer a clean alternative, yielding color-free solid polycarbonates with narrow molecular weight distributions. - Melt transesterification processes require high temperatures (up to 140 °C and above) and vacuum to drive the reaction, whereas copolymerization reactions run under elevated CO2 pressures (1 to 55 bar). - Functional polycarbonates prepared via ring-opening polymerization of azido-functionalized monomers (AzDXO) provide unique post-polymerization modification capabilities, stable up to 130 °C.

4. CONCLUSION

The analyzed patent landscape for Polycarbonate reveals three dominant technical paradigms: high-temperature continuous melt transesterification for engineering thermoplastics, transition-metal or metal-free copolymerization of CO2 with cyclic ethers for aliphatic polycarbonates, and ring-opening polymerization of functional cyclic carbonates for biomedical applications. Key process trends emphasize the optimization of catalyst selectivity (achieving >99% carbonate linkages and high regioselectivity), the elimination of metal residues to produce color-free polymers, and the precise control of molecular weight and end-group functionality (97-99% OH end groups) for polyurethane prepolymers. Gaps in the patent evidence include detailed long-term aging and mechanical performance data for the resulting copolymers.

5. REFERENCES

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